

Two-Sided Quadratic Energy in a Curved IIB Reconstruction

Stationary backgrounds, constraint reduction, boundary compatibility,
and the limits of positivity transfers

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Abstract

We study a specified bosonic realization of the first Geometric Unity action on the bundle of Lorentzian metrics over a four-dimensional base. The canonical connection metric remains curved when the observing metric is flat. An intrinsic grade-one/two distortion yields an exact real local stationary background on two certified opposite-coupling branches. Differentiating the action before restriction gives the complete mixed metric response, an algebraic distortion reduction, and the pure metric Hessian. On each branch the constraint-reduced quadratic Noether energy is unbounded above and below in both coefficient-conjugation packets. For the real packet, a finite reflection character annihilates the full fibre-integrated metric force. Compact oscillatory witnesses then satisfy every finite-order formal compatibility condition of an explicitly declared boundary polarization. This avoids the independent-metric-momentum hypothesis required by an earlier conditional acceleration argument. Bare observing-section evaluation, even augmented by the entire smooth section germ, does not carry the energy. We retain the preceding flat controls and distinguish these results from ordinary Yang–Mills, complex Chern–Simons, scalar positivity donors, and separate cutoff quantum/PDE controls. Exact coefficient calculations and five scoped Lean deductions support identified parts of the argument. The energy theorems concern a selected classical quadratic form and formal solution jets; they do not construct a closed physical evolution, a quantum spectrum, or a universal obstruction to every GU completion.

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1 Question, result, and mathematical ownership

The question is whether a proposed positive physical sector can be obtained from an action with an indefinite ambient pairing. The answer depends on its real fields, stationary background, constraints, boundary data, observation map, and Hamiltonian domain. A gauge-group designation or a positive auxiliary norm does not determine these objects.

The principal result of this manuscript is action-specific. We fix the connection metric on $Y = \text{Met}_{1,3}(X)$ and the commutator/two-symmetrized-channel realization of the first bosonic action I_1^B in Weinstein’s draft [25]. The coordinate metric, signature convention, bracket normalization, and boundary polarization below are explicit reconstruction choices. They are not claimed to be uniquely selected by that source. The source’s printed curvature residual, its square I_2^B , the observing metric, and the metric-fibre coordinate remain separate objects.

Theorem 1.1 (Main result; precise constructions below). *For the action (5), the stationary branches specified in Section 2 have the following properties.*

1. *Their imaginary coefficient-conjugation packet has a smooth, compactly supported constraint-reduced phase core on which its quadratic Noether Hamiltonian takes values tending to both infinities.*
2. *Their real packet has such families with zero metric perturbation and zero full linearized metric forcing. On a spatial torus and the reflection-invariant fibre patch of Section 6, these families satisfy all finite-order formal compatibility conditions of the odd-Dirichlet polarization in Section 4.*
3. *Every completion containing the respective core and agreeing with its quadratic energy form inherits its two-sided unboundedness. Bare smooth section-germ observation does not make the imaginary-packet Hamiltonian descend to an observed quotient.*

The proofs are given in Sections 5 to 7. The theorem neither presupposes nor constructs a global Cauchy evolution. In particular, compatible formal time jets need not converge. No Hilbert-space normalization, quantum spectral theorem, nonlinear runaway, or integrability of every linearized phase datum into nonlinear physical data is inferred. The two coupling branches are two selected values with opposite signs, not all prescribed couplings or all backgrounds.

Object	Convention and role
Carrier	Clifford-valued one-forms on Y , a base metric $g(x)$, and source gauge variables removed by exact dressing.
Real structure	A phased anti-involution slice; its real and imaginary coefficient packets are defined in (7).
Pairings	Scalar Clifford trace and exterior top-degree action pairing; reduced time symplectic form and Noether energy are distinguished.
Grading	Clifford grade, coefficient conjugation, primary time kernel, and differential order; no coefficient rank is a particle count.
Domain	Local product patches with stated variations; compact phase cores and one explicit polarization, without a closed physical generator.
Target	Semiboundedness of the selected quadratic energy and its observation descent, not every source mechanism.

The distinction between a mathematical reconstruction and a physical GU realization remains throughout. Ordinary Yang–Mills and external scalar models in Section 8 bind their own actions.

Transporting one of their conclusions would require a map of carriers and real structures, intertwining the actions, constraints, domains, and observable in question. The separate quantum/PDE controls of [Section 11](#) provide no such identification. No literature-priority claim is made for this manuscript.

2 Canonical geometry and an exact stationary background

2.1 The observing metric and the fibre metric

Let X be a four-dimensional coordinate patch and let h denote the Lorentzian symmetric matrix at a point of its metric bundle. The independent observing metric is g . Set

$$\begin{aligned} G_Y(g) &= h_{ij}dx^i dx^j + D_h(\theta, \theta), \\ D_h(A, B) &= \text{tr}(h^{-1}Ah^{-1}B) - \frac{1}{2}\text{tr}(h^{-1}A)\text{tr}(h^{-1}B), \\ \theta_{ab} &= dh_{ab} - (\Gamma(g)^c_{ia}h_{cb} + \Gamma(g)^c_{ib}h_{ac})dx^i. \end{aligned} \tag{1}$$

We use $g_* = \eta = \text{diag}(1, -1, -1, -1)$ and the resulting $(7, 7)$ ambient signature. This plus-first convention is fixed for the calculation; a source-convention bridge is not established merely by writing the same signature name. The coordinate density in symmetric fibre coordinates is

$$d\mu_Y = \mu(h)d^4x d^{10}h, \quad \mu(h) = 8|\det h|^{-2}. \tag{2}$$

It is independent of g at fixed h .

Although $\Gamma(g_*) = 0$, the coefficients depend on h . The Levi-Civita connection obtained from (1) has

$$\text{Ric}^\sharp_Y = \frac{1}{4}(P_H + P_r) - \frac{5}{4}P_0, \quad \text{Scal}(G_Y) = -10. \tag{3}$$

Here P_H is horizontal, P_0 is vertical traceless, and P_r is the vertical trace projector. Their ranks are 4, 9, 1. The vertical field $r = (0, h/2)$ satisfies

$$G_Y(r, r) = -1, \quad P_r z = -G_Y(r, z)r, \quad \nabla r = \frac{1}{4}P_H. \tag{4}$$

The coordinate Koszul calculation checks torsion, metricity, Bianchi, and these contractions [10]. For example, a vertical traceless plane has sectional curvature $-1/2$. Thus setting the spin curvature to zero would change the geometry.

Constant rescaling of g leaves $\Gamma(g)$, and hence G_Y at fixed (x, h) , unchanged. Moving $h \mapsto sh$ instead gives the familiar -8 logarithmic density derivative in fixed coordinates. That fibre motion is not an observing-metric Euler variation. This distinction corrects the interpretation of the flat controls in [Section 8.2](#).

2.2 The selected action and its real slice

Let B be the spin Levi-Civita connection of G_Y , T its dressed connection displacement, and γ Clifford multiplication with $\gamma_a\gamma_b + \gamma_b\gamma_a = 2\eta_a\delta_{ab}$ in an orthonormal frame. The selected functional is

$$I = \int_Y \text{Tr} \left[T \wedge \mathcal{S}(F_B) + \frac{1}{2}T \wedge \mathcal{S}(D_B T) + \frac{1}{3}T \wedge \mathcal{S}(T \wedge T) + \frac{\kappa}{2}T \wedge *T \right]. \tag{5}$$

The exterior integral uses scalar Clifford trace, normalized by $\text{tr}(1) = 1$. The following contraction specifies the Shiab normalization completely by linearity. For $p < q$ and Clifford coefficients C, D ,

$$\frac{\text{Tr}[e^\mu C \wedge \mathcal{S}(e^p \wedge e^q D)]}{\text{vol}_Y} = \eta_p \eta_q (\delta_{\mu p} \text{tr}(C[\gamma_q, D]) - \delta_{\mu q} \text{tr}(C[\gamma_p, D]) - \frac{\eta_\mu}{2} \text{tr}(C\{\gamma_\mu, \{\gamma_p \gamma_q, D\}\})) . \quad (6)$$

The last sign includes $\ast \text{vol}_Y = -1$ and the two factors of i in the symmetrized channels. Equation (5) is motivated by source Eq. (9.4), p. 44, but the source's displayed residual is not substituted for its Euler derivative.

Define a conjugate-linear Clifford anti-involution j by $j(\gamma_a) = -\gamma_a$ and $j(AB) = j(B)j(A)$. The source-phased real slice is $\mathcal{U} = \{Z : j(Z) = -Z\}$. Under ordinary coefficient conjugation it splits as

$$\begin{aligned} R &= \Lambda^1 + \Lambda^2 + \Lambda^5 + \Lambda^6 + \Lambda^9 + \Lambda^{10} + \Lambda^{13} + \Lambda^{14}, \\ J &= i(\Lambda^0 + \Lambda^3 + \Lambda^4 + \Lambda^7 + \Lambda^8 + \Lambda^{11} + \Lambda^{12}). \end{aligned} \quad (7)$$

All grade spaces are real before the displayed i . Commutators and i times anticommutators preserve $\mathcal{U}$; Hodge and the spin connection preserve its exterior extension. For $A, B \in \mathcal{U}$, $\overline{\text{tr}(AB)} = \text{tr}(j(AB)) = \text{tr}(BA) = \text{tr}(AB)$. The action is real on this slice. This involution is not a positive Hilbert adjoint.

2.3 Stationarity before restriction

Consider the intrinsic field

$$T_\ast(z) = a\gamma((P_H + P_0)z) + c\gamma(P_r z) - \frac{v}{8}\gamma(r)\gamma(P_H z) + w\gamma(r)\gamma(P_0 z). \quad (8)$$

The last two terms have grade two. Vary (5) in arbitrary one-form directions before inserting (8). For a test variation U , the derivative term contributes

$$\frac{1}{2} \sum_s \{ \langle U, \mathcal{S}(e^s \wedge \nabla_s T) \rangle - \langle \nabla_s T, \mathcal{S}(e^s \wedge U) \rangle \}; \quad (9)$$

the second summand is essential. The cubic derivative has both $\langle U, \mathcal{S}(T \wedge T) \rangle/3$ and $\langle T, \mathcal{S}(U \wedge T + T \wedge U) \rangle/3$. The complete distortion covector is zero exactly when the following five representative rows vanish:

$$\begin{aligned} 16E_H &= 4224a^2 + 768ac + 16a\kappa + v^2 - 72vw - 3v + 768w^2 + 108w - 84, \\ 24E_0 &= 6336a^2 + 1152ac + 24a\kappa + 3v^2 - 128vw - 12v + 896w^2 + 168w - 90, \\ 24E_r &= 7488a^2 + 24c\kappa + v^2 - 144vw - 6v + 864w^2 + 216w - 126, \\ 24E_{Hr} &= 132av - 3168aw + 12a + 4cv - 288cw - 12c + 3\kappa v, \\ 3E_{0r} &= 22av - 352aw + 3a + 2cv - 24cw - 3c - 3\kappa w. \end{aligned} \quad (10)$$

There are respectively 4, 9, 1, 4, 9 possibly nonzero rows. To see completeness, label $e^\mu \gamma_I$ by the XOR of its exterior index mask and its Clifford mask. The background has only labels 0 and {10}, where index 10 denotes r . The Clifford/exterior trace selection rule leaves 28 candidate receivers; the scalar receiver $e^{10}1$ vanishes identically. Direct evaluation checks every remaining linear, mass, and curvature receiver. Thus these are full distortion equations, not four ansatz equations.

The stationary construction [23] gives the exact algebraic branch, specified without decimal root finding. Put $a = \kappa\alpha$, $c = \kappa\beta$ and

$$\begin{aligned} D &= 22v^2 - 220vw - 15v - 3168w^2 + 732w, \\ \alpha &= -\frac{3(2v^2 - 20vw - 3v - 288w^2 - 12w)}{8D}, \\ \beta &= \frac{3(22v^2 - 220vw + 3v - 3168w^2 + 12w)}{8D}. \end{aligned} \quad (11)$$

The conic and coupling equations are

$$\begin{aligned} 3v^2 - 40vw - 512w^2 - 15v + 12w + 72 &= 0, \\ V_H R_0 - V_r H_0 &= 0, \quad \kappa^2 = -V_H/H_0, \\ H_0 &= 4224\alpha^2 + 768\alpha\beta + 16\alpha, \\ R_0 &= 7488\alpha^2 + 24\beta, \\ V_H &= v^2 - 72vw - 3v + 768w^2 + 108w - 84, \\ V_r &= v^2 - 144vw - 6v + 864w^2 + 216w - 126. \end{aligned} \quad (12)$$

For an explicit univariate specification let w_* be the unique root in

$$\frac{227427945967}{10^{12}} < w_* < \frac{227427945969}{10^{12}} \quad (13)$$

of $P(w) = \sum_{j=0}^{10} p_j w^j$, with coefficients listed in [Appendix A](#). There $v(w)$ is also given explicitly. Sturm counting certifies one root; rational intervals exclude every parameter denominator from zero and give $2.873978 < \kappa_*^2 < 2.874152$. All five rows of (10) reduce exactly to zero in $\mathbb{Q}(w_*)$, with the square root handled by $a = \kappa\alpha$ and $c = \kappa\beta$. Take $\kappa = \kappa_* > 0$ or its negative and the corresponding a, c . The map

$$(a, c, \kappa) \mapsto (-a, -c, -\kappa), \quad (v, w) \mapsto (v, w) \quad (14)$$

preserves the common zero set. Decimal values on the positive branch, for orientation only, are $(a, c, v, w, \kappa) \simeq (0.151945, -0.195077, 3.79821, 0.227428, 1.69531)$.

Proposition 2.1 (Local source-variable stationarity). *The fields (8) with the certified parameters are stationary for the selected bulk functional on a local product patch, with compact base-metric variations and compact interior distortion and gauge variations.*

Proof. Translations of x and simultaneous affine/congruence transformations of (x, h) preserve (1) at constant g . Congruence is transitive on the signature patch and transports the natural projectors and field. Consequently the exact distortion equations transport from one fibre point to the smooth patch, with local oriented spin lifts.

For the source gauge field ε , set $B_\varepsilon = \varepsilon^{-1} D_0 \varepsilon$ and $\tau = \varepsilon \varpi \varepsilon^{-1} - (D_0 \varepsilon) \varepsilon^{-1}$, where D_0 is the reference spin connection. Conjugating curvature, covariant derivative, and Shiab, and using cyclic trace, gives $I(g, \varepsilon, \varpi) = I(g, 1, \tau)$ identically. The unrestricted distortion Euler covector is zero, so the gauge-field first variation is zero.

Finally g enters the local action only through $\Gamma(g)$ and its jets. At constant g its variation contains at least one base derivative of δg ; the coefficients depend on h but not x . Integration by parts in x against compact base tests gives zero. An induced variation of $T_*(g)$ contributes zero because its distortion Euler equation is already satisfied. The fixed relatively compact fibre patch makes the variational integrals finite. No integration over the entire noncompact fibre or moving fibre boundary is used. $\square$

The coupling here is a selected background parameter, not a physical mass prediction. The failed smaller families remain informative: the three-projector grade-one ansatz forces $a = b = c = t$ and then requires both $312t^2 + \kappa t = 21/4$ and $312t^2 + \kappa t = 15/4$. In (8), setting $w = 0$ instead gives $E_0 - E_H = (v - 5/2)^2/16 + 71/64 > 0$. Neither exclusion applies to the enlarged family just constructed.

3 Coupled variation and local algebraic reduction

Use natural field coordinates $g = g_* + u$ and $T = T_*(g) + t$. This is an invertible differential change of variables, not a restriction of t to the background ansatz. It transports boundary data. The formal quadratic equations are

$$Ct + Au = 0 \quad \text{on } X \times V, \quad Mu + A^\dagger t = 0 \quad \text{on } X. \quad (15)$$

The adjoint $A^\dagger$ in this notation includes integration over V . The same $u(x)$ occurs at every fibre point; replacing its Euler equation by pointwise fibre equations or section evaluation changes the action. Here C is the actual distortion Hessian, $A = D_g E_T(g, T_*(g))$, and M is the pure-base Hessian in these coordinates.

3.1 The complete mixed operator

For a base covector q and symmetric polarization u , define

$$\begin{aligned} L_i^a{}_j &= \frac{1}{2} \eta^{ab} (q_i u_{bj} + q_j u_{bi} - q_b u_{ij}), \\ C_i(h) &= L_i^T h + h L_i, \quad k_{iA} = -D_h(C_i(h), A), \quad k_{ij} = k_{AB} = 0. \end{aligned} \quad (16)$$

With $Q = (q, 0)$, the third-base-derivative Ricci coefficient is

$$R_{ab}^{(3)} = \frac{1}{2} \left(Q_a Q^c k_{bc} + Q_b Q^c k_{ac} - Q^2 k_{ab} - Q_a Q_b \operatorname{tr}_G k \right). \quad (17)$$

All contractions here use G_Y . The second-order coefficient is

$$\delta \operatorname{Ric}_Y = \operatorname{diag}(\delta \operatorname{Ric}_g, 0_{10}). \quad (18)$$

In each case apply the grade-one Shiab covector: if $E = \operatorname{Ric}^\sharp - \operatorname{Scal} I/2$, its row on $e^\mu \gamma_\nu$ is $-E^\mu{}_\nu$. The covector array must not be confused with its Riesz endomorphism by dropping signature signs.

These formulas give $A = A_3 + A_2$, with $A_0 = A_1 = 0$. The finite second-jet calculation spans all torsion-free connection jets $N[k][i]_j^a = N[k][j]_i^a$ with closed trace $\operatorname{tr} N[k][i] = \operatorname{tr} N[i][k]$. There are 170 explicit generators spanning this 154-dimensional space. On all of them the radial variations vanish and the full formal-adjoint contribution from the moving horizontal projector cancels. Six anisotropic/sheared fibre controls supplement the coordinate covariance argument [13, 6]. The zeroth-order term vanishes because a constant g variation leaves its connection unchanged. A first connection value is generated by a coordinate change with identity first jet; naturality of the identically zero background Euler field then gives $A_1 = 0$. In particular $A(\mathcal{L}_\xi g_*) = 0$.

At the observing section $h = \eta$, $C_i = q_i u$, so k is a pure ambient metric gauge symbol and A_3 vanishes. This is not true off section. For example take $q = (1, 1, 0, 0)$ and $u = \operatorname{diag}(0, 0, 1, -1)$. Its base Ricci variation is zero. On $h = \operatorname{diag}(1 + \rho, -1, -1, -1)$ the nonzero mixed Ricci entries include

$$R_{2,02}^{(3)} = -\frac{\rho^2}{2(1 + \rho)^2}, \quad R_{3,03}^{(3)} = +\frac{\rho^2}{2(1 + \rho)^2}, \quad (19)$$

and their symmetric partners; 02, 03 denote fibre coordinates. Thus the pure natural-metric lift does not intertwine the base Ricci operator with the full ambient equation.

A leading compensator exists for $Q^2 \neq 0$. Set $d_b = Q^a k_{ab}$ and $k_\perp = k - (Q \otimes d + d \otimes Q)/Q^2$. The transverse tracefree spin-connection variation

$$y_i = \frac{1}{4} \sum_{a < b} \eta_a \eta_b (Q_b(k_\perp)_{ai} - Q_a(k_\perp)_{ib}) \gamma_a \gamma_b \quad (20)$$

satisfies $E(Q)y = A_3(q)u$, where E is the skew Euler coefficient from (9). The coefficient identity is checked on the complete transverse tracefree basis and extends by covariance and polynomiality after clearing the denominator. The unprojected spin variation has additional Euler rows and fails this check. Equation (20) cancels a principal term only; it is neither a full coupled mode nor an inverse at $Q^2 = 0$.

3.2 Two different algebraic eliminations

Ordinary coefficient conjugation fixes the action and background, so the Hessian has independent R and J packets from (7). The metric lies in R ; J has no mixed metric source at quadratic order. Clifford parity gives another decomposition, $t = o + e$, into odd and even grades. Shiab reverses parity, so

$$C = \begin{pmatrix} V_o & D_1 \\ D_1^\dagger & V_e \end{pmatrix}, \quad e = -V_e^{-1} D_1^\dagger o. \quad (21)$$

The diagonal blocks are algebraic and D_1 is first order, including connection and grade-two-background terms. For the selected background V_e is invertible: XOR labels give fourteen representative 14×14 blocks, and all 39 distinct nonconstant determinant factors have rational bounds excluding zero at (13). For example the unphased weight-one radial block has determinant

$$-\frac{\kappa(44a + 12c - 3\kappa)^{12}(528a + 40c + 3\kappa)}{1594323}. \quad (22)$$

Complexified stabilizer covariance transports determinant nonvanishing; it does not transport physical real energy signs. Phasing the real slice preserves invertibility. This elimination introduces no covector inverse, including at the null locus. The remaining odd equation is $(V_o - D_1 V_e^{-1} D_1^\dagger) o + Au = 0$. It is an equivalence of formal bulk equations, not inversion of C .

For the Hamiltonian it is more useful to eliminate the time kernel. Write the unphased real distortion operator as

$$C_R = E^\mu \nabla_\mu + V, \quad V = \kappa M_H + a H_a + c H_c + v H_v + w H_w. \quad (23)$$

Each H is the full cubic action Hessian, differentiating all three slots. The Clifford tensors are parallel, hence $\nabla E = 0$. Fix the exact time covector $q = dx^0$ on its positive-square patch.

Lemma 3.1 (Geometric primary kernel). *If $N(q)$ is a smooth frame of $\ker E(q)$, then*

$$N^T E^A N = 0, \quad N^T E^\mu \nabla_\mu N = 0. \quad (24)$$

Consequently the primary auxiliary matrix is exactly $W = N^T V N$.

Proof. Orthogonal covariance gives constant rank on the positive-square locus. Differentiate $E(q)N(q) = 0$ once and twice in covector components and multiply on the left by N^T , using skewness of $E(q)$:

$$N^T E^A N = 0, \quad N^T (E^A N_{,B} + E^B N_{,A}) = 0.$$

Since $q = dx^0$, torsion freeness makes $\nabla_A q_B$ symmetric. Contracting the second identity proves the second part of (24) in a normal frame. A change of kernel frame adds a within-kernel derivative, killed by the first identity. $\square$

Choose a smooth complement with inclusion ι and write $t = \iota s + Nz$. In the real packet put

$$F = N^T C_R \iota, \quad B_0 = N^T A, \quad z = -W^{-1}(Fs + B_0 u). \quad (25)$$

The operator F has no time derivative. All forty primary classes in each packet are invertible at the exact stationary root. Their labels are quartets under toggling indices 0 and 10; representatives have $0 \leq n_H \leq 3$, $0 \leq n_V \leq 9$, with multiplicity $\binom{3}{n_H} \binom{9}{n_V}$. Scaling even kernel coordinates by κ and dividing W by κ puts its entries in $\mathbb{Q}(w_*)$. Every determinant is nonzero there. Clifford parity P gives

$$W_- = -PW_+P, \quad P^2 = 1, \quad (26)$$

covering the mirror branch exactly. The packet coefficient counts (113792, 115584) and time-form ranks (64482, 66430) describe the raw real and imaginary coordinates; they are not physical polarization counts.

Let $D_0 = \iota^\dagger C_R \iota$ and $J_R = \iota^\dagger A$. Completing the square gives

$$\begin{aligned} I_{R,\text{red}}^{(2)} &= \frac{1}{2} \langle s, D_0 s \rangle + \langle s, J_R u \rangle + \frac{1}{2} \langle u, Mu \rangle \\ &\quad - \frac{1}{2} \langle Fs + B_0 u, W^{-1}(Fs + B_0 u) \rangle. \end{aligned} \quad (27)$$

Every adjoint ending in the base equation includes fibre integration. The remaining distortion time form is nondegenerate on its quotient. These statements determine a local primary recovery, not a closed physical domain or its boundary gauge charges.

4 The metric Hessian and the boundary qualification

4.1 Pure-base variation and acceleration

For the natural field (8) put $K = (11a + c)/2$ and define

$$F_{ij} = R(g)_{ij}^T h + h R(g)_{ij}, \quad \|F\|^2 = h^{ik} h^{jl} D_h(F_{ij}, F_{kl}).$$

This contraction is indefinite. Direct adapted-connection contraction gives

$$\begin{aligned} \text{Scal}(G_Y) &= -10 + h^{ij} \text{Ric}(g)_{ij} - \frac{1}{4} \|F\|^2, \\ \text{Ric}_Y(r, r) &= -\frac{1}{4}, \\ I(g, T_*(g)) &= \int \mu(h) \left(C_* + K h^{ij} \text{Ric}(g)_{ij} - \frac{K}{4} \|F\|^2 \right) dx dh. \end{aligned} \quad (28)$$

Here C_* is independent of g ; natural mass and cubic contractions are constant. The possible curvature correction to the first-order natural density cancels. The coordinate certificate checks the quadratic identity in all 54 curvature components and 18 coordinate-jet controls [12].

For $L_i = \delta \Gamma_i[u]$, $F_1 = \delta F$, and $Q^{ij} = \int_V \mu(h) h^{ij} dh$, the Hessian is

$$\begin{aligned} M[u, u] &= 2K \int_X Q^{ij} (L^a_{ab} L^b_{ij} - L^a_{jb} L^b_{ai}) dx \\ &\quad - \frac{K}{2} \int_{X \times V} \mu(h) \|F_1\|^2 dx dh. \end{aligned} \quad (29)$$

This includes orders two and four. For homogeneous time dependence the change $t = t' - y(h, \ddot{u})$ cancels third-time mixing. Retaining every covariant fibre and time-connection term in the zero-coordinate-time operator D_{conn} , the primary acceleration source is $N^T(A_2 - D_{\text{conn}}y - Vy)$.

The boundary successor [7] checks a complete 37-dimensional space of torsion-free homogeneous connection coefficients with $\text{tr } C_i = 0$ for $i = 1, 2, 3$. On that space

$$\begin{aligned} N^T(A_2 - D_{\text{conn}}y) &= 0, & N^T M_H y &= N^T H_a y = N^T H_c y = N^T H_v y = N^T H_w y = 0, \\ y^T M_H y &= -\|F_1\|^2/16, \\ y^T H_a y &= -\frac{52}{3}y^T M_H y, & y^T H_c y &= -\frac{4}{3}y^T M_H y, & y^T H_v y &= y^T H_w y = 0. \end{aligned} \quad (30)$$

All 1369 polarized Gram entries are checked, including cross terms. Covariance extends the identities throughout the positive-time patch. Therefore no homogeneous metric acceleration remains in the distortion primary equations, while its reduced quadratic coefficient is

$$\mathcal{K}_{\text{acc}}(u, u) = -\frac{\kappa + \frac{8}{3}(10a + c)}{16} \|F_1(u)\|^2. \quad (31)$$

At $h_0 = \text{diag}(4, -1, -1, -1)$ on the five spatial traceless polarizations,

$$\mathcal{K}_{\text{acc}} = -\frac{9}{256} \left[\kappa + \frac{8}{3}(10a + c) \right] G_F, \quad 3.083191968 < 1 + \frac{8}{3}(10\alpha + \beta) < 3.083191980, \quad (32)$$

where G_F is their positive Frobenius Gram matrix. This block stays nondegenerate after integration over a sufficiently small fibre patch.

If a boundary completion admitted these metric coordinates and velocities, primary recovery, and independently variable metric momenta, the usual canonical argument with $Q_u = \dot{u}$ would apply: its Hamiltonian contains $P_u Q_u$, unbounded in either direction at fixed nonzero Q_u . Those are substantive domain hypotheses. The following calculation shows why they cannot simply be assumed.

4.2 Raw boundary form and a declared polarization

Write $T = O + E$ by Clifford parity. Let K^μ be the raw derivative matrix in $T^T K^\mu \nabla_\mu T/2$. Its bulk coefficient is $E^\mu = (K^\mu - (K^\mu)^T)/2$, but its boundary one-form is $T^T K(n) \delta T/2$, with oriented conormal density n . For example, with $o = e^0 \gamma_0$ and $e = e^0 \gamma_0 \gamma_4$,

$$K^4(o, e) = 0, \quad K^4(e, o) = -2, \quad E^4(o, e) = 1. \quad (33)$$

Replacing K by its skew bulk coefficient would lose boundary data. Add the explicit polarization term

$$B_{\text{pol}} = -\frac{1}{2} \int_{\partial V} O^T K_{OE}(n) E. \quad (34)$$

The derivative density then has form $E^T E_{EO}^\mu \nabla_\mu O$, and its remaining boundary one-form is $E^T E_{EO}(n) \delta O$. Impose the odd natural perturbation $o = O - O_*(g) = 0$ at the fibre boundary, leaving the even perturbation free. The induced metric variation of $O_*(g)$ contributes to the base equation and is not discarded. This is a specified additional boundary choice, not a source-selected physical law. It affects lower metric orders, not (31). Because y is even, compensation preserves $o = 0$ as a boundary condition; that alone says nothing about its preservation under evolution.

4.3 The compatibility chain restricts momentum

After compensation and primary recovery write $t' = \mathcal{R}s$, where $\mathcal{R} = \iota - NW^{-1}F$ is local and contains no homogeneous metric acceleration. Define operators from the reduced action by

$$\dot{s} = \mathcal{T}s + \mathcal{U}\ddot{u}, \quad \mathcal{B}s = (P_{\text{odd}}\mathcal{R}s)|_{\partial V}. \quad (35)$$

Smooth solutions require, with $a_u = \ddot{u}$ and $j_u = \ddot{u}'$,

$$\begin{aligned} C_0 : \quad \mathcal{B}s &= 0, \\ C_1 : \quad \mathcal{B}(\mathcal{T}s + \mathcal{U}a_u) &= 0, \\ C_2 : \quad \mathcal{B}(\mathcal{T}^2s + \mathcal{T}\mathcal{U}a_u + \mathcal{U}j_u) &= 0. \end{aligned} \quad (36)$$

At $s = 0$ the original compensated even equation reads $E_{EO}(\text{dt})\dot{o} = V_{EE}ya_u$. Its five-polarization Gram is

$$y^T V_{EE} y = -\frac{9}{256} \delta G_F, \quad \delta = \kappa - \frac{52a + 4c}{3}, \quad -0.401 < \delta/\kappa < -0.399. \quad (37)$$

It is injective on that block and persists on the small patch. Thus $\mathcal{BU}$ has rank five. Nonzero acceleration with $s = 0$ fails C_1 ; at $s = a_u = 0$, nonzero jerk fails C_2 . If $\mathcal{A}$ is the integrated nondegenerate acceleration matrix, then at fixed lower jets $\delta P_u = -\mathcal{A}\delta j_u$. Consequently C_2 restricts five metric momenta at fixed distortion jets. Arbitrary independent variation of P_u does not remain in this domain.

A permanent distortion-free fibre collar is stronger still: the collar bulk equation forces $A_3 \ddot{u}' + A_2 \ddot{u} = 0$, and the independent receiver types imply $\ddot{u} = 0$ in the tested homogeneous traceless class. Neither statement establishes positive energy. They identify the limits of the earlier free-momentum proof. The real theorem below uses a different symmetry sector and does not need independently variable metric momenta.

5 Imaginary-packet energy and observation

Let $t = i\psi$ in the J packet, with real coefficients ψ . Work on $I_t \times \mathbb{R}^3 \times V$, with V sufficiently small around $h = \eta$ that dt has positive ambient square. Initial free data are smooth with compact support in $\Sigma = \mathbb{R}^3 \times V$. The packet is closed under the full Hessian by coefficient conjugation; its metric and real-packet linear sources vanish. This is a statement about the linearized equations, not nonlinear closure.

In a stationary frame write $C_R = E^0 \partial_t + C_{\text{sp}}$, retaining all connection terms. For a complement representative s set $f = N^T C_{\text{sp}} s$ and recover

$$\psi = s - NW^{-1}f. \quad (38)$$

The primary-kernel lemma and exact invertibility prove this recovery satisfies every distortion primary equation. It is a finite-order local operator and preserves support. The reduced time form is nondegenerate. The factor i reverses the quadratic action's sign:

$$\begin{aligned} L_J^{(2)} &= -\frac{1}{2} \psi^T E^0 \dot{\psi} - \frac{1}{2} \psi^T C_{\text{sp}} \psi, \\ H_{J,\text{red}}^{(2)}[s] &= \frac{1}{2} \int_{\Sigma} \left(s^T C_{\text{sp}} s - f^T W^{-1} f \right) d\mu_{\Sigma}. \end{aligned} \quad (39)$$

This is the Noether Hamiltonian, not the sign of an unreduced mass matrix. Autonomy gives its conservation on any smooth solution with vanishing boundary flux; the unboundedness argument needs only phase data.

5.1 Two exact primary columns

Choose the free direction $s_0 = e^{10}1$ in a rational orthonormal frame with radial index 10. It has a nonzero time symplectic partner: $s_0^T E^0 E^0 s_0 = -48$, and is transverse to the primary kernel. The complete primary sources and columns are

Derivative	Primary receiver	Source	Potential	Energy coefficient
e^4	$e^0 \gamma_0 \gamma_4 \gamma_{10}$	2	$+\kappa$	$-2/\kappa$
e^1	$e^0 \gamma_0 \gamma_1 \gamma_{10}$	2	$-\kappa$	$+2/\kappa$

Here e^4 is positive vertical traceless and e^1 horizontal spatial. Each of the four background cubic columns on these receivers is exactly zero, and XOR routing excludes additional primary receivers. Thus these are columns of the full inverse problem, not a truncated corner. For instance the relevant imaginary-slice terms $-2z\partial_4\psi_0 - \kappa z^2/2$ give $z = -2\partial_4\psi_0/\kappa$ and Hamiltonian coefficient $-2/\kappa$. A separate 128-state Clifford representation reproduces both columns and all their source receivers [11].

Lemma 5.1 (Compact oscillatory amplification). *Suppose the exactly reduced local quadratic form on a fixed compact support has principal amplitude term $Q(h, d\rho)(\partial_\rho a)^2$, with Q of strict constant sign and $d\rho \neq 0$ there, and all remaining terms have at most one derivative of the free amplitude. Then exact local recovery from $a_n = b \sin(n\rho)$, $b \neq 0$ smooth and compactly supported, gives*

$$H[a_n] = \frac{n^2}{2} \int Q(h, d\rho) b^2 d\mu + O(n). \quad (40)$$

Proof. Recovery differentiates the amplitude at most once. Expanding the form leaves the displayed $n^2 \cos^2(n\rho)$ term; cutoff derivatives and bounded smooth lower-order coefficients contribute at most $C_1 n + C_0$. Use $\cos^2(n\rho) = (1 + \cos(2n\rho))/2$. A vector field X with $X\rho = 1$ and integration by parts bound the oscillatory integral by $O(n^{-1})$ on the fixed support. Its contribution is therefore $O(n)$. The leading integral has the strict sign of Q . $\square$

Choose a degree-zero fibre function $y(h) = \ell(h)/h_{00}$, with $\ell(\eta) = 0$ and $dy|_\eta = e^4$. On a sufficiently small patch, $r(y) = 0$ and $|dy|^2 > 0$. Set $s_n = s_0 b \sin(ny)$ and use the complete curved recovery (38). Then

$$H_J[s_n] = -\frac{n^2}{\kappa} \int |dy|^2 b^2 d\mu_\Sigma + O(n). \quad (41)$$

For the companion use $s_n = s_0 b \sin(nx^1)$. Continuity of the exact inverse preserves its opposite principal sign on a small patch, so its energy is $+cn^2 + O(n)$ for $\kappa > 0$, with $c > 0$. The signs exchange on the mirror branch. This proves the imaginary part of Theorem 1.1; reversing the action sign also only exchanges signs. No spectral cutoff, finite-difference inference, or global evolution assumption is used.

5.2 Section visibility and failure of energy descent

Let $j_*(x) = (x, \eta)$. For $\kappa > 0$, choose b nonzero on the section in the fibre-oscillating family. Its free field vanishes there, but exact recovery gives

$$j_*^*(t_n) = -\frac{2in}{\kappa} b(x, \eta) dx^0 \gamma_0 \gamma_4 \gamma_{10}. \quad (42)$$

This is an allowed imaginary grade-three coefficient. On the negative coupling branch the spatial family supplies the analogous visible negative-energy data.

Conversely choose the fibre support disjoint from the section. Local recovery preserves that property, so the entire smooth germ at the section vanishes while the energy is nonzero. The zero field has the same observed germ and zero energy. Hence no function of that germ can equal this Hamiltonian on the core. The nondegenerate reduced time form also prevents treating these data as its presymplectic null directions. This argument applies to smooth fields. It neither substitutes for a physical reduction nor excludes nonlocal observation, integrating out fields, or a narrower admissible domain.

6 Real-packet energy with full metric decoupling

The real construction [18] uses a flat spatial three-torus, time x^0 , and a small fibre neighbourhood V of $h_0 = \text{diag}(4, -1, -1, -1)$. Choose V invariant under the three base spatial sign reflections and within the positive-time patch. The torus supplies finite volume for homogeneous base fields; compact support below is strictly inside V .

6.1 A character absent from the metric carrier

For $\epsilon_i \in \{\pm 1\}$ put

$$R_\epsilon = \text{diag}(1, \epsilon_1, \epsilon_2, \epsilon_3), \quad h \mapsto R_\epsilon^T h R_\epsilon, \quad \chi(\epsilon) = \epsilon_1 \epsilon_2 \epsilon_3. \quad (43)$$

The group $G = (\mathbb{Z}/2)^3$ acts on the base, metric fibre, and Clifford bundle. Naturality of the DeWitt metric, radial field, projectors, and connection makes it a symmetry of the chosen background and action. Its combined ambient orientation is preserved:

$$\det R_\epsilon \det(\text{Sym}^2 R_\epsilon) = (\det R_\epsilon)^6 = 1. \quad (44)$$

Thus Hodge, real structure, and time covector are preserved too. The ten homogeneous metric components transform by

$$u_{00}, u_{ii} : 1; \quad u_{0i} : \epsilon_i; \quad u_{ij} : \epsilon_i \epsilon_j \ (i \neq j). \quad (45)$$

None has character χ .

Let $P_\chi = \frac{1}{8} \sum_{g \in G} \chi(g) U_g$ on distortions. The complete mixed operator, its fibre-integrated adjoint, and the primary recovery intertwine G . It follows that

$$A^\dagger t = 0 \quad \text{whenever } P_\chi t = t \quad (46)$$

for base-homogeneous fields with arbitrary time dependence. Translation invariance makes the metric Euler expression homogeneous, so the absence of homogeneous metric receivers proves vanishing against arbitrary base metric tests, including lapse and shift. Interior support kills the induced boundary contributions. The stationary distortion operator preserves this character. Hence $u = 0$, with every time jet zero, satisfies the full linearized metric equation on this sector. This is exact decoupling, not omission of an equation. Products of two χ fields may source the trivial character, so no nonlinear truncation of this character is claimed.

Choose an equivariant smooth complement to the time kernel; finite-group averaging of an auxiliary positive coordinate form provides one. This coordinate choice is not a physical norm. With $f = N^T C_{\text{sp}} s$,

$$t = s - N W^{-1} f, \quad H_{R, \text{red}}^{(2)}[s] = -\frac{1}{2} \int \left(s^T C_{\text{sp}} s - f^T W^{-1} f \right) d\mu_\Sigma. \quad (47)$$

The sign differs from (39) because these coefficients have no common factor i . The symplectic restriction to the real χ character remains nondegenerate: distinct real characters are symplectically orthogonal and the total reduced form is nondegenerate.

6.2 The full primary inverse for two real witnesses

Let $\Gamma = \gamma_0 \gamma_1 \cdots \gamma_{13}$ and take $s_0 = e^0 \Gamma$. Its complete 18-coordinate time block satisfies $N^T s_0 = 0$ and $s_0^T E^0 E^0 s_0 = -13$. For derivatives along vertical traceless e^b , with $b = 4, 11$ and $\eta_4 = 1$, $\eta_{11} = -1$, the complete source has coefficient one on the kernel vector $r_b = e^0 \gamma_{\{0, \dots, 13\} \setminus \{b\}}$. Of the twelve target kernel directions, eleven form its connected potential block; the remaining direction decouples. On that block the exact matrices have structure

$$W_b = \begin{pmatrix} \lambda G_b & (v + 8w)h_b \\ (v + 8w)h_b^T & -\eta_b \kappa \end{pmatrix}, \quad \lambda = \kappa - \frac{116a + 4c}{3}, \quad h_b^T G_b^{-1} h_b = -\eta_b \frac{2}{33}. \quad (48)$$

Both Clifford implementations rebuild the complete derivative blocks and all five potential matrices; symbolic recovery solves every row. A Schur complement with $\tau = 116a + 4c - 3\kappa$ gives the real principal energy coefficient

$$Q_b = \frac{1}{2} f^T W^{-1} f = -\eta_b \frac{11\tau}{2[11\kappa\tau + 2(v + 8w)^2]} = -\eta_b \frac{C}{\kappa}, \quad (49)$$

where

$$C = \frac{11\kappa^2(116\alpha + 4\beta - 3)}{2[11\kappa^2(116\alpha + 4\beta - 3) + 2(v + 8w)^2]}, \quad 0.388 < C < 0.389. \quad (50)$$

The strict bounds are outward rational intervals on the exact root; numerator and denominator are separated from zero. Thus the two fibre directions have opposite signs. These coefficients depend on the background; the imaginary witness's zero cubic columns are not being generalized. On (14) they exchange signs.

6.3 Character projection preserves the energy signs

Choose a generic point near h_0 with all three h_{0i} nonzero. Its eight reflection translates are distinct. A small ball B has eight disjoint translated closures within V . Such points approach h_0 , so both strict coefficient signs persist in small neighbourhoods. For each sign choose a smooth phase ρ whose differential equals the corresponding covector at the centre; shrink B to keep it nonzero and its coefficient of strict sign. For a cutoff $b \in C_c^\infty(B)$ put $s_n = s_0 b \sin(n\rho)$, then project $\tilde{s}_n = P_\chi s_n$ and recover with the full curved operator. On B , the projected field is $s_n/8$, hence is nonzero. Locality and disjoint supports give the exact identity

$$H_R[\tilde{s}_n] = \frac{1}{8} H_R[s_n]. \quad (51)$$

Indeed all cross terms vanish and action symmetry makes the eight unscaled diagonal terms equal. Recovery commutes with the same action. The free first-order term in (47) and all lower-order recovery terms meet Lemma 5.1. Hence

$$H_R[\tilde{s}_n^-] = -c_- n^2 + O(n), \quad H_R[\tilde{s}_n^+] = +c_+ n^2 + O(n), \quad c_\pm > 0. \quad (52)$$

Their complete metric force vanishes by (46).

6.4 All finite-order formal boundary conditions

In this sector the boundary system (35) becomes $\dot{s} = \mathcal{T}s$ and $\mathcal{B}s = 0$, with $u = 0$. The reduced time-form inverse is pointwise and W^{-1} is algebraic. Thus both $\mathcal{T}$ and recovery $\mathcal{R}$ are finite-order differential operators with smooth coefficients, preserving χ . For every integer $j \geq 0$, the field $\mathcal{T}^j \tilde{s}_n$ has support in the same compact union of balls. Consequently

$$\mathcal{B}\mathcal{T}^j \tilde{s}_n = 0 \quad (j \geq 0), \quad \partial_t^j u = 0. \quad (53)$$

These formal time jets solve the differentiated coupled equations and all boundary compatibility conditions to each finite order. In particular they avoid the C_1, C_2 failure of the zero-distortion acceleration data in Section 4. This completes the real assertion of Theorem 1.1.

A differential core and compatible jets do not select a closed generator or prove convergence of its time series. The ambient ultrahyperbolic domain problem remains. The conclusion is precisely that a completion retaining this core and agreeing with its quadratic energy cannot make that form semibounded. A nonlinear interacting Hamiltonian or quantum operator is a further object requiring its own construction.

7 What a physical selection must supply

Ordinary conjugation gives the exact bosonic identity $I(g, R + J) = I(g, R - J)$. The derivative in every J direction therefore vanishes at $J = 0$, for arbitrary g, R . The real-only restriction is a consistent nonlinear bulk restriction when the boundary data respect it. Its energy is nevertheless indefinite on the core just constructed. No closure under added fermionic terms is asserted. Conversely the imaginary packet alone is not generally a nonlinear restriction: an independently checked RJJ third variation on the specified background directions equals $-124/3$ [19].

A naive ambient compact-coefficient projection is a different proposal. In the positive occupation-state matrix adjoint, $i1$ is compact, but its spatial scalar profile forces a recovered $i\gamma_0\gamma_1\gamma_{10}$ component with leading coefficient $2\partial_1\phi/\kappa$, which is noncompact in that split. The projection is not preserved by the tested primary recovery. This tested ambient split is not the source's actual $\text{Spin}(6, 4)$ shielding construction.

The observation obstruction in Section 5 also has an off-shell counterpart. A smooth bump f away from the section and the grade-one fields $U_+ = e^0\gamma_4$, $U_- = e^0\gamma_{10}$ have zero full section germ. Their wedge squares vanish; parity kills their derivative density. The curvature-linear term cancels between opposite amplitudes, leaving

$$I[\epsilon f U_\pm] + I[-\epsilon f U_\pm] - 2I[0] = \pm \kappa \epsilon^2 \int f^2 \text{vol}_Y. \quad (54)$$

This proves failure of bare action descent on the smooth carrier. It is distinct from, and weaker than, the constrained energy argument. Neither result tests a proved nonlocal physical observable or an integrating-out procedure.

A proposed physical completion must specify which fields and equations it retains, how its restriction is propagated, and how its observed energy and physical pairing are defined on one domain. A positive conserved norm could coexist with an unbounded Hamiltonian; a lower energy bound is an additional requirement. These calculations supply explicit witnesses against which that requirement can be tested. They do not show that every possible source completion, background, or physical quotient must retain the witnesses.

8 Earlier controls and external positivity proposals

These results helped formulate the physical question, but each has its own action and domain. They are not alternative derivations of the curved energy theorem. In particular an instability proved for a flat surrogate is not a growing-mode theorem for the curved background.

8.1 Positive spatial pairings do not ensure stable wave energy

For two complex scalar components on $\mathbb{R}$, fix $b > 0$ and a smooth real static function $r(x)$. Define

$$K = \begin{pmatrix} r & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ r' & 0 \end{pmatrix}, \quad U = \begin{pmatrix} 0 & -b \\ 0 & br \end{pmatrix}, \quad D = -I\partial_x^2 - B\partial_x + U. \quad (55)$$

The declared wave equation is $u_{tt} + Du = 0$, from the action $\frac{1}{2} \int (u_t^T K u_t - u_x^T K u_x + u^T \text{diag}(0, b)u)$. This is the observed two-field model, without an I1B identification [22].

Formal symmetry in a Hermitian multiplication pairing W requires $2W' = B^T W + WB$ and $WU - U^T W + W'' - (B^T W)' = 0$. Solving both equations, including an imaginary off-diagonal entry before it is forced to vanish, gives every positive solution:

$$W = s \begin{pmatrix} c^2 + \beta r + r^2/4 & \beta + r/2 \\ \beta + r/2 & 1 \end{pmatrix}, \quad s > 0, \quad c > 0, \quad |\beta| < c, \quad r'' = \frac{b}{2}(r^2 - 4c^2). \quad (56)$$

For $s = 1$, $\beta = 0$, the factor $F = \begin{pmatrix} c & 0 \\ r/2 & 1 \end{pmatrix}$ satisfies $W = F^T F$. Conjugating and applying a constant orthogonal rotation gives scalar channels

$$D_- = -\partial_x^2 + b(r/2 - c), \quad D_+ = -\partial_x^2 + b(r/2 + c). \quad (57)$$

Every other allowed pairing becomes a positive constant weight on these same channels. For bounded smooth r , the profile equation and its first integral bound the derivatives needed for this similarity, giving a closed realization from the scalar $H^2(\mathbb{R})$ domains.

Proposition 8.1 (Bounded-profile energy criterion). *Within (56) on the whole line, $D \geq 0$ precisely when $r \equiv 2c$.*

Proof. A bounded profile cannot exceed $2c$: on any component where it does, $r'' > 0$, and convexity contradicts either the endpoint values or boundedness on a half-line or the whole line. Thus $b(r/2 - c) \leq 0$. Unless $r = 2c$ identically, it is strictly negative on some interval with integral $-\delta < 0$. A trial function equal to one there and tapering to zero over lengths L has kinetic cost $2/L$ and potential at most $-\delta$. Large L and smooth approximation give a negative form value. At $r = 2c$ both scalar potentials are nonnegative. $\square$

For $k^2 = bc/2$, the exact nonconstant profile $r = 2c - 6c \text{sech}^2(kx)$ still has uniformly positive W . Its D_- eigenfunctions $\text{sech}^2(kx)$ and $\sinh(kx) \text{sech}^2(kx)$ have eigenvalues $-4k^2$ and $-k^2$. They give growing wave solutions with rates $2k$ and k on the stated domain. A positive conserved phase-space pairing cannot contain a nonzero generator eigenvector with positive real growth rate: if $Av = \gamma v$ and $A^\dagger G + GA = 0$, then $2\gamma v^\dagger G v = 0$. This is a wave statement, distinct from positivity of the spatial weight and from the original indefinite canonical action.

8.2 Flat I1B surrogates and their correction

The flat constant-coefficient completion used the same selected Shiab but set the ambient connection and curvature to zero. On fixed flat geometry, the radial restriction $T = \phi(t)\Phi_1$ omits normal bivector equations requiring $\phi' = 0$. Adding one isotropic bivector amplitude v still gives necessary equations

$$\begin{aligned}\kappa\phi + 312\phi^2 - \frac{260}{3}v^2 &= 0, \\ v' &= \frac{4}{33}v^2, \quad -11\phi' - (\kappa + \frac{568}{3}\phi)v = 0.\end{aligned}\tag{58}$$

On an interval with $v \neq 0$, differentiating the first equation forces $118976\phi^2 + 816\kappa\phi + \kappa^2 = 0$. Continuity then makes ϕ constant, contradicting the v' equation. Thus this small nonlinear restriction supplies no varying solution.

A larger linearized metric/distortion sector does have a growing mode in that surrogate. With diagonal spatial metric amplitude ψ , let (a, b, d) be time, horizontal-spatial, and vertical grade-one coefficients, and (u, w) their two bivector companions. The metric condition is $(b + 5d)'' = 0$, and the coupled characteristic determinant is

$$21600\kappa^2 z^4(113z^2 - 17\kappa^2).\tag{59}$$

For $\kappa \neq 0$ an exact mode has

$$d = e^{\gamma t}, \quad b = -5d, \quad a = 0, \quad u = -5d'/\kappa, \quad w = 7d'/\kappa, \quad \psi = 135d/(17\kappa), \quad \gamma = |\kappa|\sqrt{17/113}.\tag{60}$$

The certificate checks all ten metric rows and all 28 distortion rows in the closed label block; trace selection excludes omitted receivers. On the zero-affine branch its reduced energy is $565(d')^2/(2\kappa) - 85\kappa d^2/2$. The separate distortion-only exponent $\kappa/11$ is removed by metric coupling. Neither calculation is transferred to the curved equilibrium.

The associated co-moving flat constant-background experiment also has an exact exclusion. Its metric condition $L = 0$, together with distortion scaling $3Q + \kappa N = 0$, forces $Q = N = 0$ at nonzero κ . On diagonal grade-one fields $N = \sum t_\mu^2$. Adding the two bivector companions makes N indefinite, but the exact necessary stationary ideal, rescaled to unit coupling, is

$$\langle \partial_a(Q + N/2), \partial_b(Q + N/2), \partial_d(Q + N/2), \partial_u(Q + N/2), \partial_w(Q + N/2), N \rangle = \langle a, b, d, u, w \rangle.\tag{61}$$

This rejects that five-field nonzero constant family [8]. Its $-8L$ metric variation belongs to the declared co-moving completion; Section 2 explains why it is not the independent g variation of the canonical metric bundle. The constant-background exclusion and growing mode remain exact results on their named object, with no canonical-curved or physical GU conclusion [9].

8.3 The scope of the original objection

Nguyen–Polya’s response is dated February 23, 2021 and discusses the 2013 lecture with 2020 follow-up material as a pre-quantum classical proposal [24, pp. 1–2]. Weinstein’s written draft is dated April 1, 2021 [25]; later formulas cannot be treated as text the February authors overlooked. The response’s Section 3.1 combines a real-structure objection with an inference from complexification to nonunitarity or two-sided unbounded energy. An abstract real vector-space isomorphism is different from an identification preserving representation, real structure, and operations. Even if a real identification needs correction, complexifying notation does not itself declare all complex

components independent real physical fields. The source-phased slice in (7) is one reason those steps must be distinguished.

A charitable dynamical reading asks about a propagating Yang–Mills sector with an indefinite invariant kinetic form. That is a substantive conditional concern, but its action and physical modes need an applicability proof. The following comparator makes those assumptions explicit.

9 The Yang–Mills obstruction with its hypotheses exposed

Let G be a connected Lie group whose real Lie algebra $\mathfrak{g}$ has a nondegenerate, $\text{Ad}(G)$ -invariant symmetric bilinear form β . On $\mathbb{R}_t \times \mathbb{T}^d$, $d \geq 2$, use the flat metric of signature $(-, +, \dots, +)$, a trivial bundle, real connection components and

$$S_{\text{YM}} = -\frac{1}{4g^2} \int \beta(F_{\mu\nu}, F^{\mu\nu}) dt dx, \quad g^2 > 0. \quad (62)$$

With $E_i = F_{0i}$, the energy and Gauss constraint are

$$H = \frac{1}{2g^2} \int_{\mathbb{T}^d} \left(\sum_i \beta(E_i, E_i) + \sum_{i < j} \beta(F_{ij}, F_{ij}) \right) dx, \quad (63)$$

$$D_i E_i = 0. \quad (64)$$

Nondegeneracy of β allows the usual identification of canonical momentum with $g^{-2}\beta(E, \cdot)$. All these assumptions belong to this comparator, not to GU by default.

Proposition 9.1 (Negative energy survives the ordinary Gauss constraint). *If some $X \in \mathfrak{g}$ has $\beta(X, X) < 0$, the Hamiltonian (63) is unbounded below on smooth Gauss-compatible initial data, even after ordinary gauge equivalence. If β has both signs, the Hamiltonian is also unbounded above.*

Proof. Choose a nonzero smooth divergence-free real vector field u on the torus and take

$$A_i = 0, \quad E_i = \lambda X u_i.$$

Then $F_{ij} = 0$ and $D_i E_i = \lambda X \partial_i u_i = 0$, whereas

$$H = \frac{\lambda^2 \beta(X, X)}{2g^2} \int_{\mathbb{T}^d} |u|^2 dx \longrightarrow -\infty.$$

Gauge transformations preserve H by invariance of β , so taking the gauge quotient cannot remove these values. Using a positive direction instead proves the assertion about the upper bound. For a nonconstant transverse example one may choose $u_1 = \sin x^2$ and $u_i = 0$ for $i \neq 1$. $\square$

This is an exact classical result, not a finite-difference instability or a proof about an unspecified quantum domain. It demonstrates why the negative internal direction is not merely the removable timelike polarization of ordinary compact Yang–Mills theory. On a one-color abelian subsector, $[X, X] = 0$, so the negative sign is already visible in an ordinary transverse Maxwell mode.

For example, if a complex Lie algebra carries a nondegenerate complex symmetric invariant form b , its underlying real vector space admits

$$\beta_c(X, Y) = \text{Re}(c b(X, Y)), \quad c \neq 0.$$

It is real nondegenerate: vanishing against both Y and iY forces $b(X, Y) = 0$ for every Y . Moreover, $\beta_c(iX, iX) = -\beta_c(X, X)$, so its signature is split. This explains the kinetic-sign problem when this form and these real independent fields are actually used in (62). It says nothing about a theory with a different action or physical reduction.

The usual quantum alternative is equally concrete at the linear level. A negative transverse oscillator quantized on positive $L^2(\mathbb{R})$ has

$$\hat{H} = -\frac{1}{2}(\hat{p}^2 + \omega^2 \hat{q}^2), \quad \text{spec}(\hat{H}) = \{-\omega(n + \frac{1}{2}) : n \geq 0\}$$

in units $\hbar = 1$. It is self-adjoint and generates unitary time evolution, but is not bounded below. Exchanging the roles of annihilation and creation operators to reverse the ladder gives a negative one-particle norm if the same adjoint and canonical commutator are retained. This explains the conventional dilemma; it is not a classification of all possible contours, representations or constrained quantum theories. Witten's introductory comparison makes the corresponding action-dependent point [26, report pp. 2–3].

10 What Witten's construction does and does not say

Witten studies three-dimensional Chern–Simons theory for the complexification $G_{\mathbb{C}}$ of a compact group G , not a propagating Yang–Mills curvature-square theory. His action pairs a complex connection $\mathcal{A}$ with its complex conjugate in two Chern–Simons terms, with couplings $t = k + is$ and $\tilde{t} = k - is$. In the real- s branch relevant here, k is quantized and s is real, with $t, \tilde{t} \neq 0$ for the displayed connection formulas (for example $k = s = 1$); the physical reality condition and the choice of polarization are part of the construction. No assertion about arbitrary complex couplings is needed. The alternative imaginary- s positive construction is explicit only in genus one with restricted parameters; no all-genus positivity assertion from that branch is used.

In canonical quantization on $\Sigma \times \mathbb{R}$, with Σ a closed oriented surface, the time component of the connection imposes flatness. The phase space is a moduli space of flat complex connections, not the unconstrained space of transverse Yang–Mills waves. The Hamiltonian of this pure theory is a constraint and vanishes on reduced physical states. Thus the time evolution of this fixed, source-free closed-surface problem is the identity. A zero Hamiltonian is bounded below, but this does not imply that the state space is empty or that all topological observables are trivial. Boundary terms, puncture/source insertions and chosen boundary dynamics require a separate analysis; one may not export $H = 0$ to those problems or to a propagating theory.

The journal identity is [27]; the precise theorem used here concerns a *connection on a Hilbert bundle over Teichmüller space*, the parameter space of auxiliary complex structures on Σ . The fiber is described as $L^2(\mathcal{M}, L^k)$, where $\mathcal{M}$ is the compact-group moduli space used in the chosen polarization and L^k its quantization line bundle. For real s , the specified corrected connection δ^Q is unitary and projectively flat, with central curvature of type $(1, 1)$. The pairing has the form $\int_{\mathcal{M}} H(\chi, \psi) d\mu$, where H incorporates the regularized Laplacian determinant; conjugation by $H^{1/2}$ identifies it with the ordinary positive L^2 pairing. This is unitary *parameter-space transport*, not a theorem identifying a four-dimensional physical-time Hamiltonian. Projective flatness also retains a scalar-phase issue; it is not literal flatness.

There is a further rigor qualification. Witten gives heuristic reasoning for the quantum correction to the connection, then proves the stated properties of the selected formulas. Moduli-space singularities, noncompactness and the definition of general functional integrals are not thereby eliminated. In particular, the identification of the reduced state space uses the dense stable locus and the corresponding Narasimhan–Seshadri description; it is not an unrestricted assertion

about every singular stratum. The paper supplies a mathematical construction and a substantive counterexample to the group-only slogan in its stated quantization setting; it is not a complete axiomatic construction of interacting QFT for every complex group. The positive pairing and the closed-surface zero-Hamiltonian argument are separate ingredients. Neither should be replaced by the bare word “unitary.”

All detailed formula locators in this paragraph refer to the 44-page IASSNS-HEP-89/65 report: §3, p. 8 (closed surface); §4, pp. 19–21 (correction and theorem); §4.1, pp. 22–23, Eqs. (4.12) and (4.16) (pairings) [26]. The journal bibliographic identity is verified, but its typesetting was not independently inspected; no journal page is used as a formula locator.

10.1 Perturbative noncompact Chern–Simons and the GU transfer

Bar-Natan–Witten study perturbation theory with a noncompact gauge group and a gauge-fixed kinetic construction using an auxiliary positive form [4]. Its gauge fixing and operator are part of that construction; changing the pairing does not leave an arbitrary action and domain unchanged. Neither this perturbative result nor Witten’s parameter-space theorem identifies an interacting four-dimensional GU Hamiltonian.

In the source, I_1^B , its printed residual $\Upsilon^B = \mathcal{S}(F_A) + *\kappa T$, and $I_2^B = \|\Upsilon^B\|^2$ are separate variational owners. For a residual-square functional $S(\phi) = Q_\phi(R(\phi), R(\phi))$ at $R(\phi_0) = 0$,

$$D^2S_{\phi_0}(v, w) = 2Q_{\phi_0}(DR_{\phi_0}v, DR_{\phi_0}w).$$

An indefinite ambient form alone does not determine its restriction to the derivative image. Nor does the symbol $\|\cdot\|^2$ prove positivity. The direct I1B energy calculation in this paper differentiates its own action and solves its own primary equations; it requires neither comparator transfer.

Accordingly the group-only inference does not establish the GU-specific energy conclusion. The ordinary Yang–Mills proposition establishes its conditional obstruction on (62). The I1B result establishes a separate obstruction on (5). Citing complex Chern–Simons alone supplies no positive physical completion of either propagating model. These judgments bind the stated objects, not a critic or a theory as an undifferentiated whole.

10.2 Scalar and composite-observable donors

The dated source intake also inspected four-derivative scalar and composite-observable proposals [21]. These are independent models. In the fixed scalar basis $Q = \square\sigma$, $X = (\partial\sigma)^2$, a complete-square diagnostic for $aQ^2 + bQX + cX^2$, $a \neq 0$, is $4ac - b^2 = 0$. In the normalization $-Q^2/2 + \lambda_3XQ + \lambda_4X^2$, it becomes $\lambda_4 = -\lambda_3^2/2$, with RG tangency $\beta_4 + \lambda_3\beta_3 = 0$ on the branch. Passing that test does not prove closure of a scalar embedding under normal Euler equations.

Anderson–Bateman–Herzog–Turok’s inspected version gives three-loop renormalization identities and a nonexceptional off-shell infrared argument [2]; its on-shell inclusive probability problem is separate. The cubic momentum polynomial reduces under conservation to $2(p \cdot q)^2 - 2p^2q^2$, providing the soft-momentum factors needed in that model. A transfer must retain that interaction grammar, not only a conformal resemblance. There is also a bounded displayed-coefficient correction: from

$$L = -\Upsilon\square\Omega - \frac{g}{6}\Omega^2\Upsilon^2$$

one obtains $\Upsilon = -3\square\Omega/(g\Omega^2)$ and $L_{\text{eff}} = 3(\square\Omega/\Omega)^2/(2g)$ for $g\Omega \neq 0$. The inspected Eq. (5.4) instead prints $1/(2g)$. The source intake checks the PDF, HTML, and original TeX identity. This

discrepancy does not negate the separate loop calculation. Bateman–Turok’s tree-level ghost-parity result [5] does not by itself establish loop positivity for a GU sector.

Amaral–Lemes’ inspected second version constructs a cocycle in an enlarged background-equivariant theory and a positive leading one-loop composite spectral density [1]. BRST closure and positivity of the full interacting operator are different conclusions. Asorey and collaborators analyze a subtracted composite spectral representation and a bubble-resummed bound-state correlator of conjugate-mass fields [3]. Continuum density, isolated pole residue, and contact subtractions must be checked separately; this is not positivity of every Schwinger function. These exact-version source records provide possible methods and tests, without a native GU transfer or a claim about newer literature.

11 Separate cutoff quantum and gauge–matter controls

The merged research corpus includes a distinct variational and PDE sequence. This section states its current yield with its own objects; none is used as a premise of [Theorem 1.1](#). Take the normalized three-torus, cube cutoff $K_N = \{k \in \mathbb{Z}^3 : \|k\|_\infty \leq N\}$, fixed $m, g > 0$, and $\omega_k = (m^2 + |k|^2)^{1/2}$. The positive Wick-square control has free Gaussian form q_0 and

$$C_N = \sum_{k \in K_N} (2\omega_k)^{-1}, \quad W_N = \int_{\mathbb{T}^3} (\phi_N^2 - 3C_N)^2, \quad E_N = \inf_{\|\psi\|=1} \{q_0[\psi] + g\mathbb{E}_{|\psi|^2} W_N\}. \quad (65)$$

This positive shifted polynomial convention must not be confused with an unshifted normal-ordered quartic Hamiltonian.

11.1 From sign texture to the quartic exponent

For a real degree- N field f , set $\delta = \int (f^2 - 1)^2$ and let a fixed-ratio Fourier shell of $\text{sgn } f$ carry L^2 mass at least $\eta > 0$. Since $\|f - \text{sgn } f\|_2^2 \leq \delta$, a small defect forces $\|\nabla f\|_2^2 \geq \alpha^2 \eta N^2/4$. On $B = \{|f| < 1/2\}$, its measure is at most $16\delta/9$ and Bernstein–Hölder gives gradient cost at most $CN^2\sqrt{\delta}$. On B^c , differentiating $f^2 - 1$ bounds it by $4N^2\delta$. After canceling N^2 ,

$$\alpha^2 \eta/4 \leq C\sqrt{\delta} + 4\delta.$$

Thus the defect has a strictly positive cutoff-independent lower bound under the fixed-shell hypothesis. A high-frequency sine has defect $3/8$ and sign-shell mass $8/\pi^2$, so the exponent zero is sharp. The probability-level lifting of this estimate, combined with the low-shell Fisher bound, proves

$$E_N = \Theta_g(N^4). \quad (66)$$

The vacuum supplies the matching upper bound $6gC_N^2$. The earlier lamellar exclusions and $N^{-4/3}$ defect estimate remain valid but are superseded, on the same fixed-shell problem, by this stronger result [17]. Equation (66) identifies an exponent, not the unrestricted coefficient or a bounded-error recentering.

11.2 A class coefficient and an unrestricted descent

The cube geometry matters:

$$C_N/N^2 \longrightarrow c_C = 12 \log \frac{1 + \sqrt{3}}{\sqrt{2}} - \pi.$$

For a continuum squeeze profile s on $Q = [-1, 1]^3$, define

$$A[s] = \frac{1}{2} \int_Q \frac{s(x)}{|x|} dx, \quad F[s] = \frac{1}{4} \int_Q |x|(s + s^{-1} - 2) dx.$$

At fixed variance the unique free-cost minimizer is $s_\kappa(x) = |x|/\sqrt{|x|^2 + \kappa}$. After optimizing the constant mean, the stationary diagonal Gaussian coefficient is the minimum of $F_*(A) + 6gA(2c_C - A)$. Its positive parameter solves $\kappa_g = 24g(c_C - A(\kappa_g))$. This improves the uniform-squeeze class for every $g > 0$. Finite-cutoff profiling supplies a global minimum within that Gaussian class, denoted λ_N^{prof} [16]. Moment-matched Gaussian comparison extends the same minimum to the translation-invariant finite-Fisher class with nonnegative integrated Wick defect $\int (4h\mu_3 + \mu_4 - 3v^2) \geq 0$ [15].

The sign condition is restrictive. At the Gaussian minimum, the residual has precisely third and fourth Gaussian chaos:

$$\begin{aligned} (H_N - \lambda_N^{\text{prof}})G_N &= R_N G_N, \\ R_N &= g \int \left[: \eta^4 :_{\Gamma_N} + 4h_N : \eta^3 :_{\Gamma_N} \right], \\ \sigma_N^2 &= g^2 \left(96h_N^2 \int \Gamma_N^3 + 24 \int \Gamma_N^4 \right). \end{aligned} \tag{67}$$

Positive Fourier convolution counts give $\sigma_N = \Theta_g(N^{5/2})$. The two-dimensional Ritz matrix on G_N and $R_N G_N / \sigma_N$ has energy drop

$$\Delta_N = \frac{\sqrt{\delta_N^2 + 4\sigma_N^2} - \delta_N}{2}, \quad |\delta_N| \leq C_g \sigma_N, \quad \Delta_N = \Theta_g(N^{5/2}).$$

Thus the profiled Gaussian misses the unrestricted translation-invariant bottom by at least this divergent amount. Its value plus an $o(N^{5/2})$ shift cannot recenter that bottom to bounded error. The drop is still $o(N^4)$ and does not decide the unrestricted leading coefficient [20].

The latest result covers every finite-second-moment location law π on the zero-mode coordinate with the same profiled covariance S_N . Fisher convexity and the moment lower bound have exact gap

$$U - L = \frac{\omega_0 \text{Var}_\pi A}{s_{N,0}(s_{N,0} + \text{Var}_\pi A)} \leq \frac{\omega_0}{s_{N,0}} = O_g(N).$$

Linearity of potential expectation then yields

$$\lambda_N^{\text{prof}} - O_g(N) \leq \inf_\pi Q_N(\pi) \leq \lambda_N^{\text{prof}}.$$

Arbitrarily many zero-mode locations with this common covariance retain the same leading class coefficient. Covariance heterogeneity, spatial textures, and the Ritz direction above are outside the theorem [14].

11.3 Exact cancellation without a global dispersive theorem

The separate temporal-gauge Maxwell–matter line studies opposite charges $\pm e$ on a torus. Let $a(t)$ be harmonic holonomy, $M = \sum_k (|\phi_{+,k}|^2 + |\phi_{-,k}|^2)$, and $D = \sum_k k(|\phi_{+,k}|^2 - |\phi_{-,k}|^2)$. The exact bare Fourier energy satisfies

$$E_0 = H_a + ea \cdot D - \frac{1}{2}e^2|a|^2 M, \quad E'_0 = ea \cdot D' - \frac{1}{2}e^2|a|^2 M'. \tag{68}$$

It removes $\dot{a}$ from this linear harmonic calculation, but for $m > 0$ the bound still spends $2|e||a| + (e^2/m)|a|^2$. Earlier current and radial normal forms, covariant integration by parts, and moving analytic radii must retain their remaining budgets. Static flat connections show that a raw potential-size radius cost is not always dynamically necessary. Time-dependent periodic controls show that positive conserved energy and Gauss neutrality alone need not make harmonic electric total variation finite.

For a nonzero lattice vector k perpendicular to the holonomy plane,

$$a(t) = A(\cos \Omega t, \sin \Omega t, 0), \quad \phi_+ = \phi_- = Re^{i\omega t} e^{ik \cdot x}, \quad (69)$$

with $\Omega = \sqrt{2}eR$ and $\omega^2 = m^2 + |k|^2 + e^2 A^2$, solves the stated coupled control. Opposite Gauss densities cancel; the currents add to $-2e^2 R^2 a$. It has no homogeneous matter mode, but $|\dot{a}|$ is nonzero constant. Therefore excluding only $k = 0$ does not produce dispersion. If the two frequencies are commensurate the full fields are periodic; in general the holonomy is periodic and the fields are bounded quasiperiodic. The energy and nonintegrable-variation conclusions hold in either case. This does not reject scattering under stronger localization or nonresonance hypotheses [14].

None of (66)–(69) supplies a source-owned continuum Hamiltonian, interacting BRST closure, physical quotient, or a global evolution theorem for the curved I1B action.

12 Evidence, reproducibility, and remaining physical questions

The derivations have three distinct forms of support. Exact finite calculations determine coefficient matrices, root isolation, determinant nonvanishing, and rational sign bounds. Analytic arguments use naturality, symmetry, compact support, integration by parts, and local differential recovery to obtain the continuum statements. A small Lean module proves five explicitly conditional deductions. These layers are complementary; a successful finite script is not itself a proof of a global domain theorem.

The sparse Clifford implementation evaluates exterior/Clifford products and differentiated action rows. The independent implementation realizes fourteen gamma matrices on the 128 occupation states of seven fermionic modes: positive generators are creation plus annihilation, negative ones creation minus annihilation. It checks the Clifford relations on every state and uses normalized matrix trace. It imports neither the sparse engine nor a coefficient bank for the designated witness calculations. This corroborates the finite energy inputs and selected Gram entries; it is not an outside review of the analytic proofs.

The module `Lean/GUFormalization/NguyenPositivityKernels.lean` proves: a witness refutes a universal failure predicate; a positive conserved form excludes a supplied growing eigenvector; equal observation images with a nonzero symmetric action difference prevent descent; and the specified necessary three-block polynomial system has no real solution, including its zero-coupling case. The first deduction has no axioms; the others report only `propext`, `Classical.choice`, and `Quot.sound`. An external exact probe checks the action-to-polynomial coefficients. Neither the full geometry and variational calculus nor Witten’s construction nor the curved energy theorems have been formalized by that module.

All sixteen local Python certificates and the named Lean build passed in the corpus integration at revision `f18b19f6`. That integration also reconciled the upstream quantum/PDE sequence and retained the existing contribution contracts. Earlier receipt hashes pin the text verified at their own dates; later scope updates do not retroactively change those runs. The evidence map and integration receipt are `explorations/nguyen-gu-critique/README.md` and `lab/process/na`

`tive-i1b-corpus-integration.json`. The manuscript’s typesetting receipt separately checks compilation, references, glyphs, and layout; it certifies no mathematics.

12.1 The physical question after the energy calculation

The direct result blocks semiboundedness of the specified quadratic form on any domain that retains the constructed cores. The real restriction, ordinary local boundary compatibility in the declared polarization, overall sign reversal, and bare smooth observation do not remove that result. The following objects remain unconstructed: a source-selected physical admissibility rule, its propagated coupled constraint domain, an observed energy map on that domain, and an interacting quantum Hamiltonian with its physical pairing.

A future admissibility rule must identify precisely which witnesses it excludes and why the intended dynamics preserve that exclusion. A different background or source-supported action is a different mathematical problem, not a sign change of the present theorem. Fermionic closure, nonlinear evolution, and the source’s actual shielding mechanism also require separate derivations. In repository terms, LT-SM8 and LT-GR6b remain **NEEDS**; the physical BV/BRST pairing condition **INHERITANCE_BRIDGE** remains undischarged; SC-META-53 remains **UNCERTAIN**. These labels report scope, not additional hypotheses or proof steps. No canon, source polarity, prediction credit, or publication status is advanced by this manuscript.

A Exact algebraic background specification

The polynomial used in (13) is specified by the following integer coefficients, in descending order. Together with (11)–(12), these make the background independent of rounded numerical constants.

j	p_j in $P(w) = \sum p_j w^j$
10	1078253165818470400
9	3877334434099568640
8	4643792466602204160
7	1707092071119812736
6	-454780741544188128
5	-354008474391384144
4	-41983168422902724
3	1153971394741872
2	7933181509779615
1	1617316736755599
0	-494757479546640

The associated rational function is $v(w) = N_v(w)/D_v(w)$, where

$$\begin{aligned}
N_v(w) &= -96471654400w^6 - 210114455040w^5 - 29233525248w^4 \\
&\quad + 98962215264w^3 + 12067809624w^2 - 6319064340w - 617509656, \\
D_v(w) &= 12058956800w^5 + 28525361280w^4 + 18854889984w^3 \\
&\quad + 3709530864w^2 + 545539698w - 749271789.
\end{aligned}$$

The stationary certificate derives this linear-remainder formula, checks its denominators are coprime to P , isolates the root by Sturm counting, and reduces all five Euler equations exactly. The complete primary matrices in Section 3 are rebuilt from the action rather than inferred from their numerical eigenvalues.

B Companion derivations and executable checks

The following filenames are relative to `tests/channel-swings/`. Each Python probe runs with SymPy 1.14.0; the exact mathematical scope belongs to its linked derivation and receipt, not to the count of checks.

Certificate	Principal checked inputs
<code>quantum_positivity_scalar_transfer_probe.py</code>	Scalar discriminant, soft-momentum identity, auxiliary normalization, and closure countercontrol.
<code>nguyen_static_pairing_energy_probe.py</code>	Full Hermitian symmetrizer, channel similarity, and exact growing modes.
<code>native_i1b_scalar_closure_probe.py</code>	Flat normal equations, all-row coupled mode, and nonzero spatial momentum control.
<code>native_i1b_constant_background_probe.py</code>	Co-moving density, full polynomial coefficients, and exact stationary ideal.
<code>native_zorro_geometry_probe.py</code>	Koszul geometry, curvature, projectors, metricity, torsion, and Bianchi.
<code>native_zorro_i1b_probe.py</code>	Spin-curvature/Shiab contraction, full action receiver equations, and failed three-block family.
<code>native_zorro_stationary_background_probe.py</code>	Five full Euler rows, degree-ten branch, denominator bounds, and root isolation.
<code>native_zorro_metric_perturbation_probe.py</code>	Mixed metric jets, section/off-section controls, and principal compensation.
<code>native_zorro_algebraic_reduction_probe.py</code>	Complete mixed coefficient, gauge-symbol identities, and fourteen even potentials.
<code>native_zorro_constrained_energy_probe.py</code>	Forty imaginary primary classes, mirror congruence, and both energy coefficients.
<code>native_zorro_matrix_energy_probe.py</code>	Independent occupation-state Clifford algebra and complete imaginary witness blocks.
<code>native_zorro_real_packet_probe.py</code>	Forty real primary classes, interaction witness, and ambient compact-recovery counterexample.
<code>native_zorro_real_metric_probe.py</code>	Pure-base Hessian, compensator terms, and five-polarization acceleration matrix.
<code>native_zorro_boundary_constraint_probe.py</code>	Complete homogeneous Gram identities, raw boundary form, and rank-five factors.
<code>native_zorro_real_energy_probe.py</code>	Both real inverse blocks, rational signs, orientation, character absence, and eight-point orbit.
<code>nguyen_lean_coefficients_probe.py</code>	Five action-to-polynomial coefficients supplied to the scoped Lean deductions.

From the repository root, the existing harness discovers the native certificates, including those

reused as helpers:

```
uv run --no-project --with sympy==1.14.0 python -B \
  scripts/reproduce_all.py --quick --tracked-only \
  -k native_zorro --timeout 900
```

Repeat with `-k native_i1b` for the flat controls. The three remaining scalar/pairing/coefficient probes may be run directly under the same pinned environment. The formal target is `lake build GUFormalization.NguyenPositivityKernels`. No full repository test sweep is implied by these scoped commands.

C Coverage of the related quantum/PDE sequence

The following groups index the complete merged K1538–K1600 sequence. They share neither action ownership nor a physical domain with the main I1B theorem. Their dated derivations under `explorations/` and `explorations/conditional-build/`, exact receipts under `lab/process/`, and paired certificates under `tests/channel-swings/` remain their research owners.

Group	Contribution and retained limit
K1538–1542	Amplitude rigidity, ultraviolet shell covariance demand, exclusion of globally coherent sign cones, and the sign-texture boundary. An order- N^2 trial was still an open target at this stage.
K1543–1547	Exact square-wave truncation tails and lamellar interaction/shell estimates; exclusion is specific to that family.
K1548–1553	Phase balance, coarea/Bernstein defect bound, localized Fejer control, mixture lift, and superquadratic ground energy. The quantitative bound is strengthened by the next group.
K1554–1559	Fixed-shell positive defect gap, sharp sine control, quartic ground-energy exponent, and gauge-potential normal-form exchange; no unrestricted leading coefficient or continuum limit.
K1560–1565	Cube Weyl constants, optimized uniform squeezing, radial normal form, and residual-gauge current identity. The variational coefficient is class-specific.
K1566–1570	Profiled covariance improves the Gaussian coefficient; covariant integration by parts controls a differentiated-current term. Remaining analytic budgets persist.
K1571–1575	Coupling asymptotics, exact finite-cutoff Gaussian minimum, moving-radius absorption, and a weight-ratio limitation.
K1576–1580	Matching coefficient for nonnegative Wick-defect stationary laws; static flat-connection controls expose avoidable raw potential-size radius loss.
K1581–1585	Strict residual descent outside the Gaussian class, negative-defect Fisher direction, and harmonic/oscillatory holonomy split. Integrability remains conditional.
K1586–1590	Quantified $N^{5/2}$ Ritz drop and failure of Gaussian bounded-error recentring; periodic harmonic-electric control.
K1591–1595	Symmetric profiled-cat Fisher sandwich, coefficient rigidity, opposite-charge angular cancellation, and analytic-tier conservation on the chosen orbit.
K1596–1600	Arbitrary zero-mode location mixtures over one common covariance, unequal-mode bare-energy normal form, and transverse nonzero-mode bounded periodic/quasiperiodic controls. General mixing, continuum limits, and global scattering remain outside these results.

References

- [1] M. M. Amaral and V. E. R. Lemes. Background-equivariant BRST observables and i-particle propagators from an auxiliary quartet in $SU(3)$ yang-mills. <https://arxiv.org/abs/2605.16172v2>, 2026. Inspected version 2, October 5, 2026.
- [2] Maegan Anderson, Sam Bateman, Franz Herzog, and Neil Turok. On divergences in a four-derivative scalar field theory. <https://arxiv.org/abs/2608.12210v1>, 2026. Inspected version 1, August 12, 2026; source custody in the October 7 repository intake.
- [3] Manuel Asorey, Gastão Krein, Miguel Pardina, and Ilya L. Shapiro. Reflection positivity in a higher-derivative model with physical bound states of ghosts. <https://arxiv.org/abs/2511.15283v1>, 2025. Inspected version 1, November 19, 2025.
- [4] Dror Bar-Natan and Edward Witten. Perturbative expansion of Chern–Simons theory with non-compact gauge group. *Communications in Mathematical Physics*, 141:423–440, 1991. Inspected author preprint IASSNS-HEP-91/4, January 1991, 29 pages. <https://www.math.toronto.edu/~drorbn/papers/NonCompact/NonCompact.pdf>. References to sections 2–4 and equations use this preprint.
- [5] Sam Bateman and Neil Turok. Escape from ostrogradsky via hidden ghost parity. <https://arxiv.org/abs/2607.00096v1>, 2026. Inspected version 1, June 30, 2026.
- [6] Complete mixed derivative and algebraic distortion reduction on the curved I1B branch, 2026. Unpublished companion research record. `explorations/conditional-build/native-zorro-algebraic-reduction-2026-10-08.md`.
- [7] Real I1B boundary constraints and the limit of the free-momentum argument, 2026. Unpublished companion research record. `explorations/conditional-build/native-zorro-boundary-constraints-2026-10-08.md`.
- [8] Constant-background obstruction in the selected flat I1B germ, 2026. Unpublished companion research record. `explorations/conditional-build/native-i1b-constant-background-2026-10-08.md`.
- [9] Scalar closure and a growing mode in the selected flat I1B germ, 2026. Unpublished companion research record. `explorations/conditional-build/native-i1b-scalar-closure-2026-10-08.md`.
- [10] The strongest objection, assessed end to end, 2026. Unpublished companion research record. `explorations/nguyen-gu-critique/section-3-1-end-to-end-assessment-2026-10-08.md`.
- [11] A two-sided constraint-compatible energy obstruction on the stationary curved I1B branches, 2026. Unpublished companion research record. `explorations/conditional-build/native-zorro-constrained-energy-2026-10-08.md`.
- [12] Pure metric hessian and a conditional real acceleration obstruction, 2026. Unpublished companion research record. `explorations/conditional-build/native-zorro-real-metric-hessian-2026-10-08.md`.
- [13] The observing-metric response and a leading distortion compensator, 2026. Unpublished companion research record. `explorations/conditional-build/native-zorro-metric-perturbation-2026-10-08.md`.
- [14] Location-mixture rigidity and transverse nonzero-mode cancellation, k1596–k1600, 2026. Unpublished companion research record. `explorations/k1596-k1600-location-mixtures-transverse-modes-2026-10-09.md`.
- [15] Wick-dominant non-gaussian coefficient and flat-connection radius boundary, k1576–k1580, 2026. Unpublished companion research record. `explorations/k1576-k1580-nongaussian-coefficient-flat-connection-2026-10-09.md`.
- [16] Profiled gaussian coefficient and differentiated-current boundary, k1566–k1570, 2026. Unpublished companion research record. `explorations/k1566-k1570-profile-current-boundary-2026-10-09.md`.

- [17] Spectral shell gap and gauge normal-form boundary, k1554–k1559, 2026. Unpublished companion research record. `explorations/conditional-build/k1554-k1559-spectral-shell-and-normal-form-boundary-2026-10-08.md`.
- [18] Two-sided real IIB energy on a metric-decoupled compact phase core, 2026. Unpublished companion research record. `explorations/conditional-build/native-zorro-real-energy-2026-10-09.md`.
- [19] Real-packet primary reduction and the physical-selection boundary of curved IIB, 2026. Unpublished companion research record. `explorations/conditional-build/native-zorro-real-packet-and-selection-2026-10-08.md`.
- [20] Profiled residual ritz scale and harmonic-electric periodic obstruction, k1586–k1590, 2026. Unpublished companion research record. `explorations/k1586-k1590-residual-ritz-harmonic-orbit-2026-10-09.md`.
- [21] Primary papers for quantum positivity and scalar sector closure, 2026. Unpublished companion research record. `lab/sources/quantum-positivity-primary-pack-2026-10-07.md`.
- [22] Positive pairings and unstable waves in the observed two-field model, 2026. Unpublished companion research record. `explorations/conditional-build/nguyen-static-pairing-and-energy-2026-10-07.md`.
- [23] A real stationary curved background for the selected canonical IIB reconstruction, 2026. Unpublished companion research record. `explorations/conditional-build/native-zorro-stationary-background-2026-10-08.md`.
- [24] Timothy Nguyen and Theo Polya. A response to geometric unity. https://files.timothynguyen.org/geometric_unity.pdf, 2021. February 23, 2021. Especially sections 2.3, 3.1 and 3.4, pp. 5–6, 8.
- [25] Eric R. Weinstein. Geometric unity: Author’s working draft, v. 1.0. https://geometricunity.nyc3.digitaloceanspaces.com/Geometric_Unity-Draft-April-1st-2021.pdf, 2021. April 1, 2021. Source-native formulas and proposals, not a completed quantum theory.
- [26] Edward Witten. Quantization of Chern–Simons gauge theory with complex gauge group. Technical Report IASSNS-HEP-89/65, Institute for Advanced Study, December 1989. Inspected 44-page report, KEK scan 90-09-085. <https://lib-extopc.kek.jp/preprints/PDF/1990/9009/9009085.pdf>. Report page numbers are not journal page numbers.
- [27] Edward Witten. Quantization of Chern–Simons gauge theory with complex gauge group. *Communications in Mathematical Physics*, 137:29–66, 1991. Journal bibliographic identity. Detailed inspected locators in this note refer to the separately listed report.